Yifei YUAN

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EDUCATION

The University of Hong Kong Hong Kong
PhD in Civil Engineering 12/2026 – 12/2030
• Supervisor: Prof. CHEN Ji

The Chinese University of Hong Kong Hong Kong
MPhil in Geography and Resource Management (Unfinished) 08/2026 – 10/2026
BSSc in Economics (first class honours) 09/2021 – 07/2026
Semester Exchange at University of Geneva, Switzerland 02/2024 – 07/2024

RESEARCH INTEREST

My research interests center on geospatial technology, environmental impact assessment, and environmental economics. My current research focuses on the following themes:

- Remote sensing applications in social computing
- Global-scale assessment of next-generation photovoltaic development
- Life cycle assessment modeling for sustainability analysis

PUBLICATIONS

[1] **Yuan, Y.**, Sun, L. & Chen, J. Solar poverty alleviation program raises local incomes and lowers air pollution in rural China. ***Communications Sustainability.*** 1(1), 87 (2026). https://doi.org/10.1038/s44458-026-00091-1 **(**Selected as **Featured Article)**

[2] Zhao, Y., Zhu, Z., Jia, M., Liu, Y., **Yuan, Y.**, Sun, L. Migration of global mangrove greenness centroids and regional contributions (In Chinese). (2026) ***National Remote Sensing Bulletin《遥感学报》***

[3] **Yuan, Y.**, Lee, HF., Zeng, X. *et al.* Global assessment of perovskite tandem retrofits for solar farms. (2026) (**Under Review** at ***Nature Sustainability***).

[4] Zeng, X., Sun, W., **Yuan, Y.** *et al.* Technology-resolved mapping reveals lifecycle outcomes of global utility-scale photovoltaics. (2026) (**Under Review** at ***Nature Sustainability***).

[5] Shen, Y., Sun, L., **Yuan, Y.** *et al.* Global solar PV deployment lags in regions where favorable siting conditions coincide with electricity-access deficits. (2026) (**Under Review** at ***The Innovation***).

[6] **Yuan, Y.**, Zhang, Y., Zhao, Y. *et al.* Vegetation responses to photovoltaic power stations and their methodological interpretation across land systems. (2026) (Submitted).

[7] **Invention Patent:** A Unified Organization and Fusion Management Method and System for Life Cycle Assessment Data. Granted

[8] **Invention patent:** Document Parsing and Confidence Audit Method Based on Constrained Reasoning of Life Cycle Knowledge Graph. Granted

HONOURS & AWARDS

[1] HKU Postgraduate Scholarship
[2] CUHK Full Scholarship (Gaokao admission track, Hubei Province)
[3] HKSAR Government Scholarship 2022/23, 2023/24, 2025/2026
[4] Dean's List 2022/2023, 2023/2024, 2025/2026
[5] Asia-Pacific Economic Cooperation Scholarship
[6] Yasumoto International Exchange Scholarship
[7] S.H. Ho Foundation Scholarship
[8] S.H. Ho College Advanced Scholarship
[9] S.H. Ho College Matriculation Scholarships for Academic Excellence

RESEARCH EXPERIENCE

Chinese Academy of Sciences Shenzhen
Shenzhen Institutes of Advanced Technology 12/2024 – 08/2026
Research Intern (Supervisor: Prof. SUN Liqun)

- Leveraged ArcGIS and Python to conduct spatiotemporal analysis of global urban nighttime light dynamics, quantifying significant shifts in urban activity patterns across major regions worldwide.
- Participated in the life-cycle assessment and carbon accounting module for a national-level lithium supply chain governance project. Modeled the full carbon footprint across 5 mining-to-battery production facilities in Jiangxi Province, identifying key emission reduction nodes and delivering data-driven recommendations for policy formulation.
- Built a comprehensive panel dataset by integrating multi-source remote sensing and socio-economic data. Implemented econometric models to identify and quantify the causal effects of environmental policies on regional development.

The Chinese University of Hong Kong Hong Kong
Department of Economics 09/2025 – 12/2025
Dissertation Project (Supervisor: Prof. PEI Guangyu)

- Built a county-level panel dataset using ArcGIS and Google Earth Engine based on remote sensing data.
- Visualized the energy structure of impoverished counties with Python and constructed PSM-DID models via Stata.
- Empirically verified the significant role of photovoltaic systems in SO_2 emission reduction and revealed the heterogeneous impacts across regions with different income levels and industrial dependence.

CUHK Business School Hong Kong
School of Accountancy 08/2024 – 09/2024
Research Assistant (Supervisor: Prof. ZHOU Yuqing)

- Cleaned and managed economics dataset. Participated in construction of a matched financial dataset of U.S. publicly listed companies and their global subsidiaries, preparing them for analysis and empirical modeling.
- Coordinated with research collaborators and supported research administration tasks, including organizing datasets, literature summaries.

PROFESSIONAL EXPERIENCE

ByteDance Shenzhen
Data Analyst Intern 09/2023 – 12/2023

- Conducted research and produced data analysis reports in Southeast Asia market. Provided strategies for operations.
- Participated in e-commerce campaigns and managed weekly operation. Updated strategies based on market report.

EasyTransfer Beijing
Global Market Intern 05/2023 – 07/2023

- Conducted market research in Hong Kong and produced 2 reports about the market size of Hong Kong.
- Participated in market expansion activity and conducted community operational data analysis.

EXTRACURRICULAR ACTIVITIES

Ronald McDonald's House Charities 09/2025 – 05/2026

- Designed and organized weekend retreat workshop series for sick children and their families, incorporating hands-on craft activities and orientation games to boost engagement and provide emotional support.

Mainland Undergraduate Association Orientation Camp 08/2022 – 10/2022

- Participated in the design of a series of orientation activities to help new students integrate quickly and collected and analyzed feedback to provide insights for activity experience optimization.

SKILLS & INTERESTS

Language: English (fluent), Mandarin (native)
Software: ArcGIS, STATA, Python, R, OpenLCA